MARIYAM AMIN

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ACADEMIC BACKGROUND

BSc in Biomedical Engineering *2023–2027*

University of Engineering and Technology *Lahore, Pakistan*

CGPA: 3.62/4 (6 semesters)

RESEARCH EXPERIENCE

Ihya Lab for Mechatronics & Resuscitation | Research Intern *2026*

Project: Biomedical Signal Acquisition & Analysis *UET, Lahore*

– Performed real-time acquisition, processing, and analysis of **4 biomedical signals** (ECG, PPG, spirometry, and EMG) at sampling rates up to 1,000 Hz.

– Tools Used: NI ELVIS II+, iWorx sensors, MATLAB, NI-DAQmx, Signal Processing Toolbox

Project: Gait Assessment using Motion Capture System

– Assisted in 3D gait assessment using synchronized motion-capture cameras and post-processing of gait data, analyzing parameters such as stride length, step length, cadence, gait velocity, and joint angles.

– Tools Used: QTM/QDS, Optical MoCap, Marker-Based Kinematics, L-Bar/Wand Calibration, MATLAB

Mayo Hospital | Research Assistant *2026 –Present*

Supervised by Dr. Amna Riaz

Project: Comparative Assessment of Immersive and Non-Immersive Rehabilitation Using Wearable Multimodal Sensing

– Working on EMG and IMU data acquisition using Shimmer sensors, and analysis to assess upper-limb motor function and rehabilitation outcomes in post-stroke rehabilitation.

An Adaptive PT^{++} Based ECG Arrhythmia Detection & Clinical Risk Stratification System | Research Paper

To be published

– Developed an adaptive ECG arrhythmia detection framework integrating Pan-Tompkins++ and cardiac feature extraction, achieving 95.83% accuracy, 94.59% sensitivity, and 100% specificity on the MIT-BIH Arrhythmia Database.

INDUSTRIAL EXPERIENCE

Biomedical Intern | Sheikh Zayed Hospital *2024*

– Assisted with 20+ medical devices across 4+ departments, supporting equipment operation, clinical workflow optimization.

ACADEMIC PROJECTS

- **Wearable Upper-Limb Exoskeleton for Biomechanical Motion Assistance | Final Year Project** *Ongoing*

  Developing a wearable exoskeleton for upper-limb motion assistance, focusing on joint actuation & ergonomic alignment.

- **Vision-Based Real-Time Prosthetic Hand Control Using Computer Vision** *2026*

  Implemented gesture recognition to map movements to prosthetic actuation, achieving 94% accuracy with <150 ms latency.

  – *Tools Used:* Python, OpenCV, MediaPipe, Arduino Uno, MG996R servos, Fusion 360

- **Computational Modeling & Simulation of Ultrasound Imaging** *2026*

  Modeled ultrasound propagation, acoustic fields, RF signals, and image reconstruction using simulated transducer arrays.

  – *Tools Used:* MATLAB, SIMUS/MUST, phased-array and linear-array transducers

- **Biomechanical Modeling of Upper & Lower Limb Kinematics** *2025*

  Analyzed muscle forces, external loading, moment arms, and joint torque during simulated flexion–extension.

  – *Tools Used:* PASCO Structures, force sensor, data acquisition, force–time analysis

- **Design of a Home-Based Wearable Monitoring System for Heart Failure Patients | PBL** *2025*

  Designed a wearable cardiovascular monitoring architecture integrating arterial pressure and single-lead ECG acquisition.

• **CRISPR-Cas9 Gene Editing Strategies in Glioblastoma | Review Article** *2025*

Co-authored a review of CRISPR-Cas9 oncogene silencing, immune enhancement, and blood-brain-barrier delivery strategies for glioblastoma therapy.

• **Surface EMG Signal Acquisition & Muscle Activity Analysis** *2024*

Acquired and filtered surface EMG signals to isolate muscle activation patterns for physiological analysis.

– *Tools Used:* EMG electrodes, signal filtering, biomedical instrumentation, signal acquisition

• **IMU-Based Real-Time Posture Detection & Feedback System** *2024*

Developed C7–T1 IMU-based posture monitoring detecting >30° spinal deviation sustained for 5s with corrective feedback.

– *Tools Used:* Arduino Uno, MPU6050, accelerometer, gyroscope, servo motors

SKILLS

Programming Languages & Tools: MATLAB, Arduino, VS Code, Fusion 360, Proteus, Material Studio; Machine Learning, Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Imbalanced-learn), C++, NotebookLM

Bioinformatics Tools: NCBI, UniProt, Protein Data Bank (PDB), KEGG Pathway Database

Relevant Coursework: Biomechanics, Biomedical Instrumentation, Microprocessor Systems, Biomedical Electronics, Biomaterials, Signals & Systems

Soft Skills: Problem-Solving, Research, Project Management, Teamwork, Communication, Analytical Thinking

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

• **Vice President | Literary & Debating Society, UET, New Campus** *June 2026–Present*

– Led inter-university literary and debating competitions with 300+ registrations and 20+ participating institutions across Pakistan.

• **Co-Founder | Muhandis** *2026*

– Co-founded a STEM education initiative delivering hands-on learning experiences to underprivileged students, with a focus on empowering young girls to pursue engineering fields.

• **Campus Director | Hult Prize, UET Lahore** *2024–2025*

– Selected as Campus Director; became the first woman to hold the position in UET's history, with the event recognized among the Top 3 entrepreneurial events nationally.

• **Student Ambassador | Alkhidmat Foundation** *2024–Present*

– Coordinated Punjab's largest MedCon with 1,500+ participants, represented the organization across 10+ initiatives, and spearheaded flood-relief fundraising campaigns.

CERTIFICATES & FELLOWSHIPS

• AI/ML Fellowship Program (ACM) *Remote*

• Prompt Engineering Research & Integration Internship (Excelerate) *Remote*

• Bioinformatics Remote Internship (CodeAlpha) *Remote*

• Robotics, AI & Industry 4.0 Innovation Lab Workshop (Rochester Institute of Technology)

• Machine Learning for Healthcare Applications (Coursera)

• MATLAB On-Ramp (MathWorks)

• Effective Leadership within a Nutrition Response (UNICEF)

HONORS & ACHIEVEMENTS

• **President's Merit Laptop:** Awarded by the Government of Pakistan for academic excellence.

• **Award of Excellence:** Recognized by the Campus Coordinator for outstanding contribution to campus initiatives.

• **Project Exhibition**: Secured 1st Prize at the Biomedical Engineering Project Exhibition.

• **Scholarships**: HSSC Merit-Based Scholarship (PKR 200,000); STEP(MDCAT) Scholarship; SSC Merit-Based Scholarship for ranking among the Top 3 students in high school.